



Spider Emergency Room





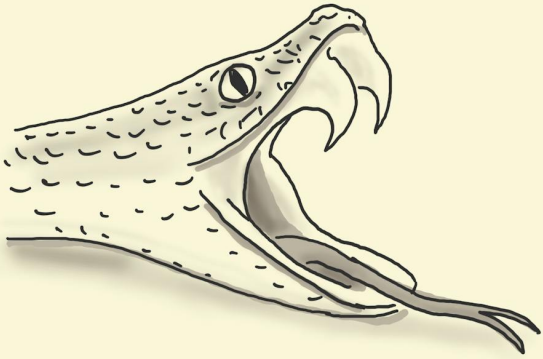
01

Poisonous Spiders 101



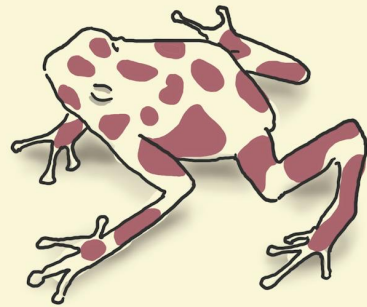
Venomous or Poisonous?

WHAT'S THE DIFFERENCE BETWEEN VENOMOUS AND POISONOUS



VENOM INJECTED
THROUGH BITES,
STINGS & THE LIKE

Snakes, spiders,
stingrays...



POISON ABSORBED
CONSUMED OR
INHALED

Amphibians, plants,
fungi...

Introduction to Spider Venom

While most of the spiders are venomous, only a small fraction of the 50,000 known spider species are medically significant to humans. Venom is a specialized form of poison delivered through fangs, primarily used for hunting and defense.

Venom vs. Poison

Venom is injected into the victim, while poison is absorbed or ingested. This distinction is crucial for understanding the mechanisms of spider bites and their effects on humans.

Medical Relevance

While many spiders have venom, only a few pose a significant medical risk. Understanding which spiders are dangerous helps in diagnosing and treating spider bites effectively.

Venom Types & Human Impact

Neurotoxic Venom

Neurotoxic venom attacks the nervous system, causing symptoms like muscle cramps, sweating, and intense pain. It can lead to systemic effects and is often associated with spiders like the Black Widow.



Necrotic and Irritant Venom

Necrotic venom destroys tissue, leading to ulcers and slow-healing wounds. Irritant hairs, like those of the New World Tarantula, cause skin and respiratory reactions but are not venomous.

02

The Big Five

Black Widow: Neurotoxic Cramp Master

Appearance

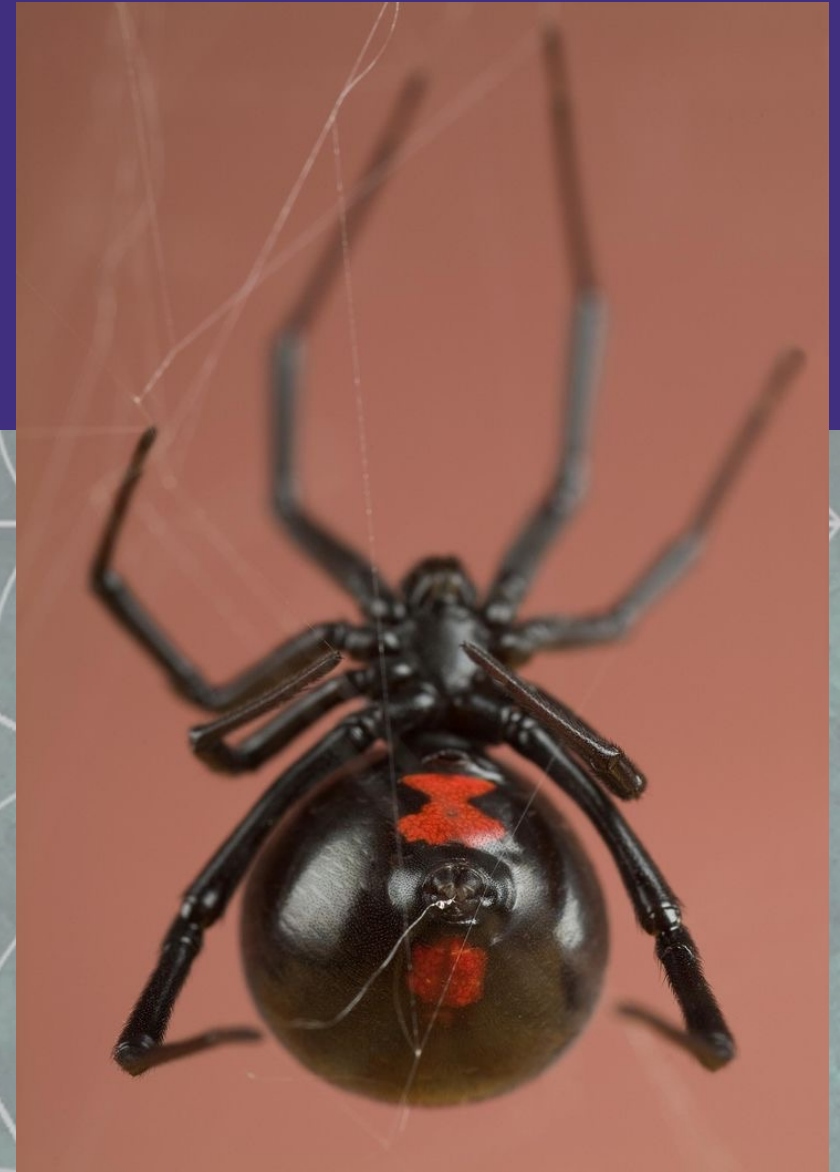
The Black Widow is easily identified by its glossy black body and red hourglass marking. It is found worldwide in dark, undisturbed areas like woodsheds and garages.

Venom Effects

Its neurotoxic venom causes severe muscle cramps, sweating, and hypertension within an hour of the bite. Symptoms are systemic but rarely fatal.

Bite Frequency

Bites are rare and usually occur when the spider is provoked. Understanding the habitat and behavior helps in diagnosing Black Widow bites.



Sydney Funnel-web: Lightning Neurotoxin



Appearance and Habitat

The Sydney Funnel-web is an aggressive spider found in eastern Australia, particularly in gardens and burrows. It has large fangs and a glossy black body.

Venom Effects

Its highly neurotoxic venom causes rapid onset of symptoms within minutes, including drooling, twitching, and confusion. Antivenom is crucial for treatment.

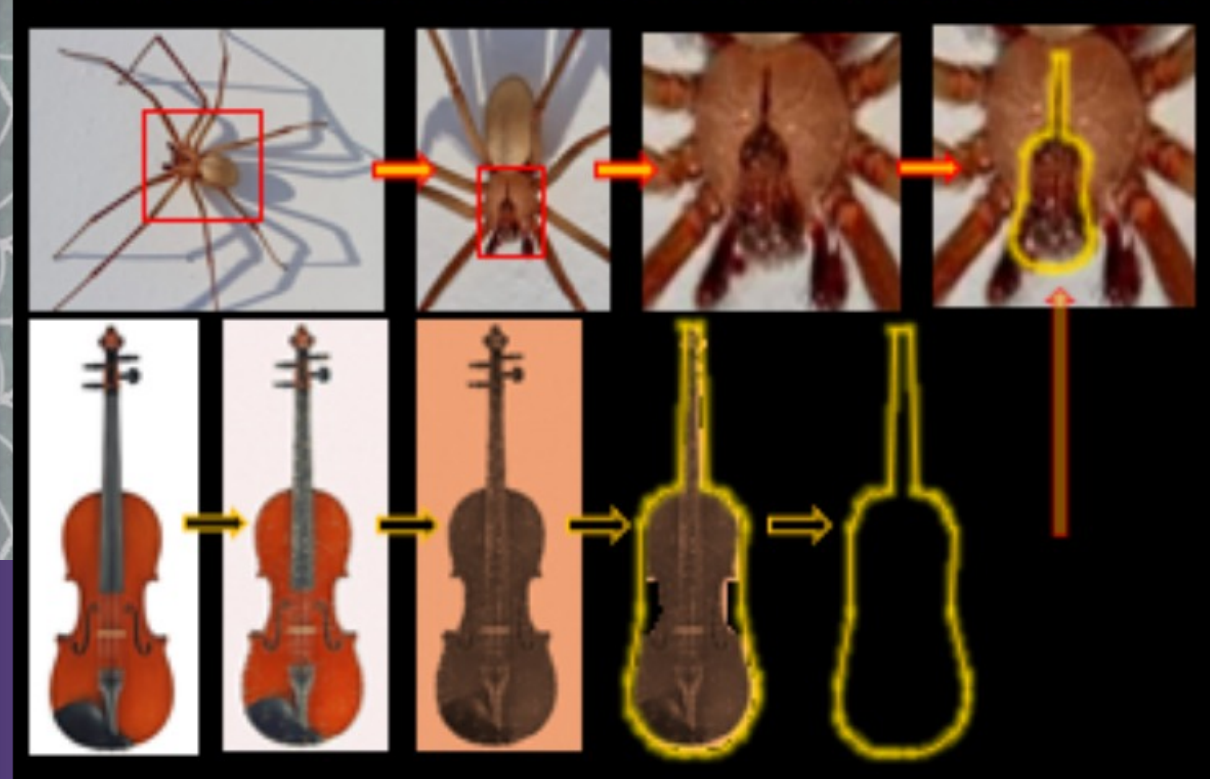
Brown Recluse: Necrotic Ulcer Maker

Appearance

The Brown Recluse has a violin-shaped mark on its body and is found in quiet indoor spaces like closets and garages in North America.

Venom Effects

The bite is initially painless but develops into a bullseye lesion with a dark center and surrounding hard tissue, leading to a necrotic ulcer over days.



Wolf Spider: Painful but Harmless

Appearance and Habitat

Wolf spiders are ground-dwelling hunters with striped bodies, found worldwide in gardens and garages. Their bite is painful but causes only local redness and swelling, with no systemic effects.



New World Tarantula: Hair Kicks

Defense Mechanism

New World Tarantulas defend themselves by flicking urticating hairs, not biting. These hairs embed in the skin and airways, causing burning rashes and breathing difficulty.



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Spider Emergency Room

Patient A: Rash Woes

Patient Presentation

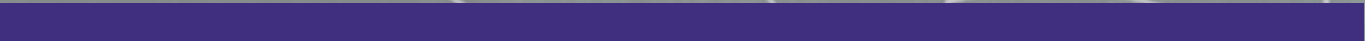
Patient A felt a storm of tiny needles while cleaning her pet spider's glass tank, followed by a burning, itchy rash and mild breathing difficulty.





Reference Table for Patient A

Spider	Venom/Defense Type	Typical Human Symptoms
Black Widow	Neurotoxic	Affects nervous system. Causes severe muscle cramps, sweating, intense pain, rapid onset. Non-necrotic.
Sydney Funnel-web	Highly Neurotoxic (to primates)	Extremely rapid onset. Causes tingling, muscle spasms, profuse salivation (drooling), disorientation, extreme pain. Medical emergency.
New World Tarantula	Urticating Hairs (Defense, not venom)	Barbed hairs are kicked off the abdomen. Cause intense irritation, rash, and potential respiratory distress if inhaled. Not a bite.
Wolf Spider	Mildly Venomous	Bite is mechanically painful (big fangs). Venom causes only local pain and swelling, not systemic or necrotic effects.
Brown Recluse	Cytotoxic / Necrotic	Initially painless. After hours/days, causes necrosis (tissue death) and a characteristic slow-healing ulcer.



Solution: Urticating Hairs



Correct Diagnosis

Patient A was affected by New World Tarantula urticating hairs, not a bite. The rash and breathing difficulty match this spider's defensive mechanism.

Treatment

Pain can be treated with ice and analgesics. If the exposure occurred on a limb, elevating the limb can sometimes help. Itching can be treated with medicine like ibuprofen. Removal of urticating hairs using tape or candle wax has been reported to reduce pain. Wounds should be monitored for signs of secondary infection, but antibiotics are not routinely recommended for tarantula bites or hair exposure.

Patient B: Cramping & Sweating

Patient Presentation

Patient B experienced severe abdominal cramps, heavy sweating, and high blood pressure. Symptoms began about an hour after cleaning a woodshed.

Bite Description

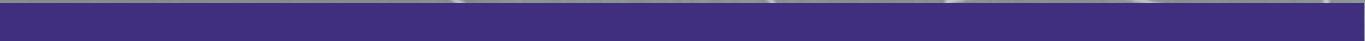
The bite site shows two tiny puncture marks. The patient felt fine initially but developed symptoms within an hour, indicating a neurotoxic reaction.





Reference Table for Patient B

Spider	Venom/Defense Type	Typical Human Symptoms
Black Widow	Neurotoxic	Affects nervous system. Causes severe muscle cramps, sweating, intense pain, rapid onset. Non-necrotic.
Sydney Funnel-web	Highly Neurotoxic (to primates)	Extremely rapid onset. Causes tingling, muscle spasms, profuse salivation (drooling), disorientation, extreme pain. Medical emergency.
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Brown Recluse	Cytotoxic / Necrotic	Initially painless. After hours/days, causes necrosis (tissue death) and a characteristic slow-healing ulcer.



Solution: Black Widow Bite

Correct Diagnosis

Patient B was bitten by a Black Widow. The woodshed habitat, delayed onset, and systemic neurotoxic symptoms match the Black Widow's profile.



Treatment

- Wash the area with soap and water.
- Apply an ice pack to help reduce pain and swelling and to delay the effects of the venom.
- Elevate the wound site, if possible.
- Don't attempt to remove the venom (sucking out the venom is a dangerous practice that's just as likely to envenom the person trying to do it).
- Seek medical care as soon as possible

Patient C: Pinprick Pain, No Illness



Patient Presentation

Patient C felt a sharp pinprick while putting on a shoe left in the garage. The bite is painful and swollen but without systemic symptoms.



Reference Table for Patient C

Spider	Venom/Defense Type	Typical Human Symptoms
Black Widow	Neurotoxic	Affects nervous system. Causes severe muscle cramps, sweating, intense pain, rapid onset. Non-necrotic.
Sydney Funnel-web	Highly Neurotoxic (to primates)	Extremely rapid onset. Causes tingling, muscle spasms, profuse salivation (drooling), disorientation, extreme pain. Medical emergency.
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Wolf Spider	Mildly Venomous	Bite is mechanically painful (big fangs). Venom causes only local pain and swelling, not systemic or necrotic effects.
Brown Recluse	Cytotoxic / Necrotic	Initially painless. After hours/days, causes necrosis (tissue death) and a characteristic slow-healing ulcer.

Solution: Wolf Spider

Correct Diagnosis

Patient C was bitten by a Wolf Spider. The localized pain and swelling without systemic effects match this spider's profile.

Treatment

Clean the bite with warm water and soap.
Apply an ice pack or a cold, damp washcloth to the bite.
Take an antihistamine to relieve itching. You want to avoid scratching, which can increase your risk of infection.
Take an over-the-counter (OTC) pain reliever to ease pain and swelling.



Patient D: Twitching & Drooling

Patient Presentation

Patient D was bitten on the leg while gardening and now exhibits rapid onset of severe neurotoxic symptoms, including intense pain, disorientation, drooling, and muscle twitching.

Bite Description

The bite occurred 15 minutes ago, and symptoms developed frighteningly fast. The rapid onset suggests a highly neurotoxic spider.

Medical Emergency

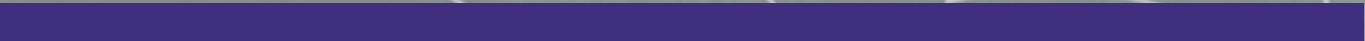
The severity and speed of symptoms indicate a medical emergency. Quick identification and treatment are crucial for patient D.





Reference Table for Patient D

Spider	Venom/Defense Type	Typical Human Symptoms
Black Widow	Neurotoxic	Affects nervous system. Causes severe muscle cramps, sweating, intense pain, rapid onset. Non-necrotic.
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Brown Recluse	Cytotoxic / Necrotic	Initially painless. After hours/days, causes necrosis (tissue death) and a characteristic slow-healing ulcer.



Solution: Sydney Funnel-web

Correct Diagnosis

Patient D was bitten by a Sydney Funnel-web. The rapid onset of severe neurotoxic symptoms, including drooling and twitching, matches this spider's profile.

Treatment

- Call the medical professionals immediately.
- Use a bandage or any fabric available and wrap it firmly around the affected area, starting from the bite site and working towards the extremities.
- Restrict movement to prevent the venom from spreading rapidly through the bloodstream.
- Keep a close eye on the victim's vital signs, including breathing, pulse, and consciousness and be prepared to administer CPR if necessary.

NSW boy survives funnel-web spider bite after taking 12 vials of anti-venom

 By 9NEWS | 11:07am Feb 24, 2017



A funnel-web spider; Matthew Mitchell was bitten on his finger by a spider hidden in a shoe. (AAP/Supplied)

Patient E: Painless Bite, Dark Ulcer

Patient Presentation

Patient E noticed no pain yesterday but now sees a bullseye lesion on the thigh turning dark and hard with an emerging ulcer.

Bite Description

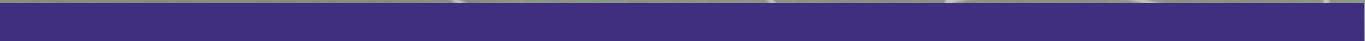
The initial bite was painless, but the lesion developed over days, indicating a necrotic reaction.





Reference Table for Patient E

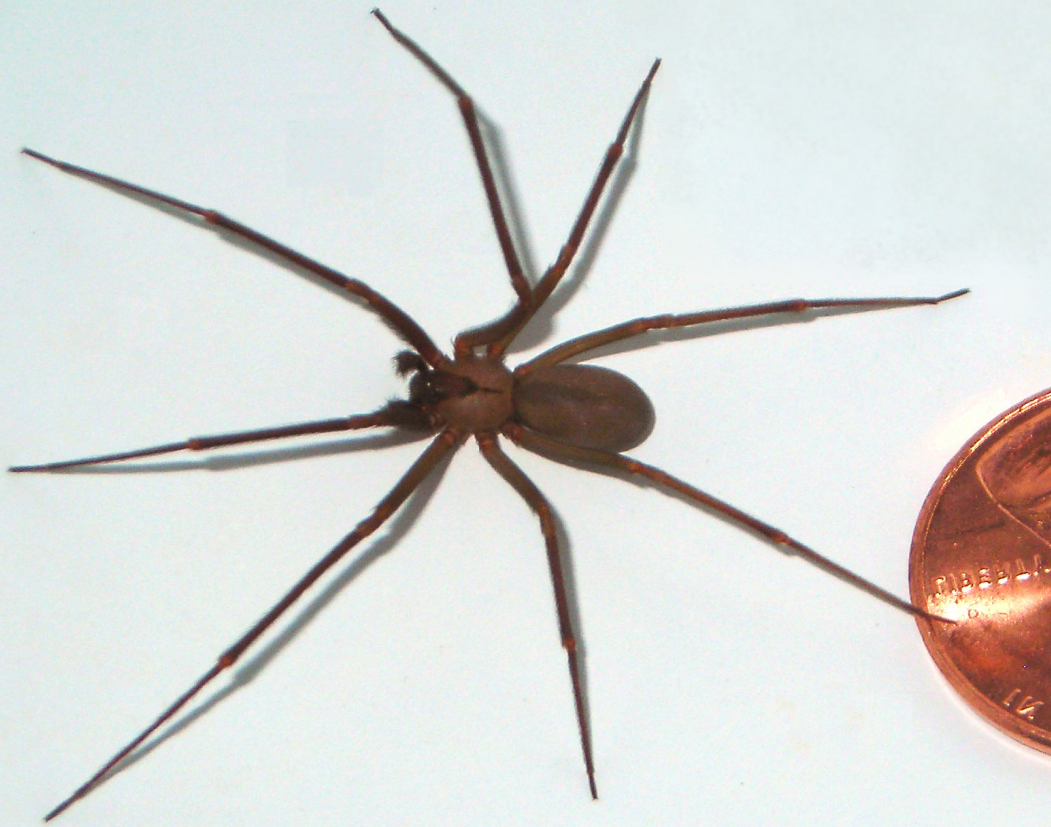
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Brown Recluse	Cytotoxic / Necrotic	Initially painless. After hours/days, causes necrosis (tissue death) and a characteristic slow-healing ulcer.



Solution: Brown Recluse

Correct Diagnosis

Patient E was bitten by a Brown Recluse. The painless initial bite followed by a necrotic ulcer matches this spider's profile.



Treatment

- Gently wash the bite with soap and water.
- Use a cold, wet cloth or an ice pack wrapped in a cloth for 10–20 minutes at a time to help with pain and swelling.
- If the bite is on an arm or leg, keep it elevated to help reduce swelling.
- Do not touch or scratch the bite.
- Keep a close eye on the bite for signs of infection or increasing pain.
- Get emergency care if you experience dizziness, trouble breathing, or a severe reaction.



THANK YOU

